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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

Semester - II University Examination, May 2010

B.Tech. (CE/EE/IT)

EE - 101 Fundamentals of Electrical & Electronics Engineering

Date: 15.05.2010, Saturday Time: 12:30p.m to 3:30p.m Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary.

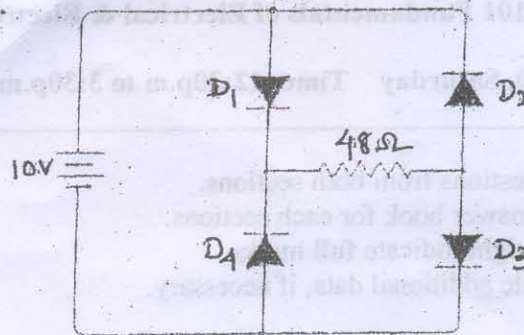
SECTION I

- Q-1 A. Write technical notes on : (1) Hysteresis loss (2) Eddy current loss 7
- B. An iron ring of mean diameter 35 cm, with cross section area $3.5 \text{ cm} \times 3.3 \text{ cm}$ is wound with coil of 455 turns. The flux in the ring is 1.1 mWb, when a current through a coil 1.7 amp. Find (1) Reluctance of Magnetic Circuit (2) Relative permeability of material (3) mmf required to establish flux of 1.0 mWb. 3
- C. Distinguish between (1) Dynamically induced emf (2) Statically induced emf. 2
- Q-2 A. Derive an expression for equivalent resistance of a number of resistances connected in (1) Series (2) Parallel 7
- B. The resistance of a given conductor is 60 ohm at 20°C . Find its temperature coefficient and resistance at 30°C . Take $\alpha_0 = 0.0043 \text{ }^\circ\text{C}^{-1}$. 3
- C. Why electrical appliances are always connected in parallel not in series? 2
- OR
- Q-2 A. Obtain the expression for the capacitance of a parallel plate capacitor with different dielectric materials. 7
- B. A capacitor of $50 \mu\text{F}$ is connected through 100 kilo ohms resistance to 230 V dc supply. Calculate the time taken to reach the capacitor voltage to 200 V after closure of switch. 2
- C. A capacitor is made of two plates with an area of 11 cm^2 which are separated by a Mica sheet 2 mm thick. If relative permittivity of Mica is 6, find its capacitance. If now one plate is moved further to give an air gap 0.5 mm wide between the plate and mica, find the change in capacitance. 3
- Q-3 A. Derive an expression for the instantaneous value of alternating sinusoidal emf in terms of its maximum value, angular frequency and time. 7
- B. Three currents i_1 , i_2 , i_3 are expressed by the expression $10 \sin(314t + 0^\circ)$, $20 \sin(314t + \pi/12)$ and $10 \sin(314t + \pi/2)$ respectively are added together. Find (1) Expression of total current. (2) RMS value of total current (3) Frequency. 4
- OR
- Q-3 A. Derive an expression of current and power factor for R-L-C series circuit connected to single phase a.c. supply. Also draw vector diagram for the circuit with (1) $X_L > X_C$, (2) $X_L < X_C$ and (3) $X_L = X_C$. 7
- B. Prove analytically that the power factor of a purely resistive circuit is unity. And also derive an expression for power. 4

[P.T.O.]

SECTION - II

- Q-4 A. Draw and explain the $V-I$ characteristics of PN junction. Also explain the operation of half-wave rectifier. 5
- B. Explain the operation of Negative Clamper. 3
- C. Calculate current through 48Ω resistor in the below circuit. Assume diodes to be of silicon and forward resistance of each diode is 2Ω . 4



- Q-5 A. How Zener diode is work as Voltage Regulator. Also explain any two filters circuits in detail. 7
- B. Explain following Logic Gates in detail (i) NOR gate; (ii) AND gate. 2
- C. Give various classifications of computers? 2

OR

- Q-5 A. (I) Explain any three terms
(a) rise time, (b) fall time, (c) periodic signal, (d) duty cycle 7
- (II) (a) Perform $(5174)_8 \cdot (7425)_8$ using 8's complement.
(b) Perform $(11001)_2 - (01110)_2$
- B. Covert given Gray code to Binary code 2
(i) 110011, (ii) 111011
- C. (i) Convert $(875)_{10}$ number into octal number. 2
(ii) Convert $(11011010010111000)_2$ number into Hexa-decimal number.

- Q-6 A. Explain: how transistor uses as an amplifier? Derive the relationship between α and β . 7
- B. Give definition of α , also derive the output current equation in Common Base configuration. 3
- C. Only draw input and output characteristic curves of Common Emitter configuration 2

OR

- Q-6 A. Explain in detail input and output characteristic curves of Common Base configuration. 7
- B. How will you draw DC load line on the output characteristic of a transistor? What is its importance? 2
- C. A transistor is connected to a CE configuration where collector supply voltage is 8V and voltage drop across resistor R_C is 0.8V. Take $\alpha=0.96$ and find 3
(i) collector-emitter voltage
(ii) base current

